Assessing the impact of particle size of mixed household waste on syngas production: a hybrid analysis

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01 INTRODUCTION

India has a complicated waste management problem. It regularly deals with heterogenous mixed-waste streams, which are often difficult to separate. Gasification provides a promising route for conversion of these wastes to useful products; however, the desired output strongly depends on feedstock characteristics. Understanding the trends in the composition of resultant syngas could aid in formulating effective pre-treatment strategies for chemical recycling of such waste.

02 OBJECTIVES

- ✓ To determine the changes in syngas composition with change in average particle size
- ✓ To measure differences in the composition of Syngas with each sample produces using a 0-D model
- ✓ To use established mechanism sets for modelling pyrolysis and gasification processes happening in a batch pyro-gasifier

03 METHODOLOGY

Plywood sawdust, Kraft paper, and black LDPE garbage bags were used as representative feedstocks for mixed household waste, of which plywood sawdust was sieved and divided into mixtures. Proximate analysis was performed on the mixtures and C,H,O values were obtained using empirical correlations. Extractive content was calculated and fed into a model built on Python using Cantera, a chemical and thermodynamic simulation library.

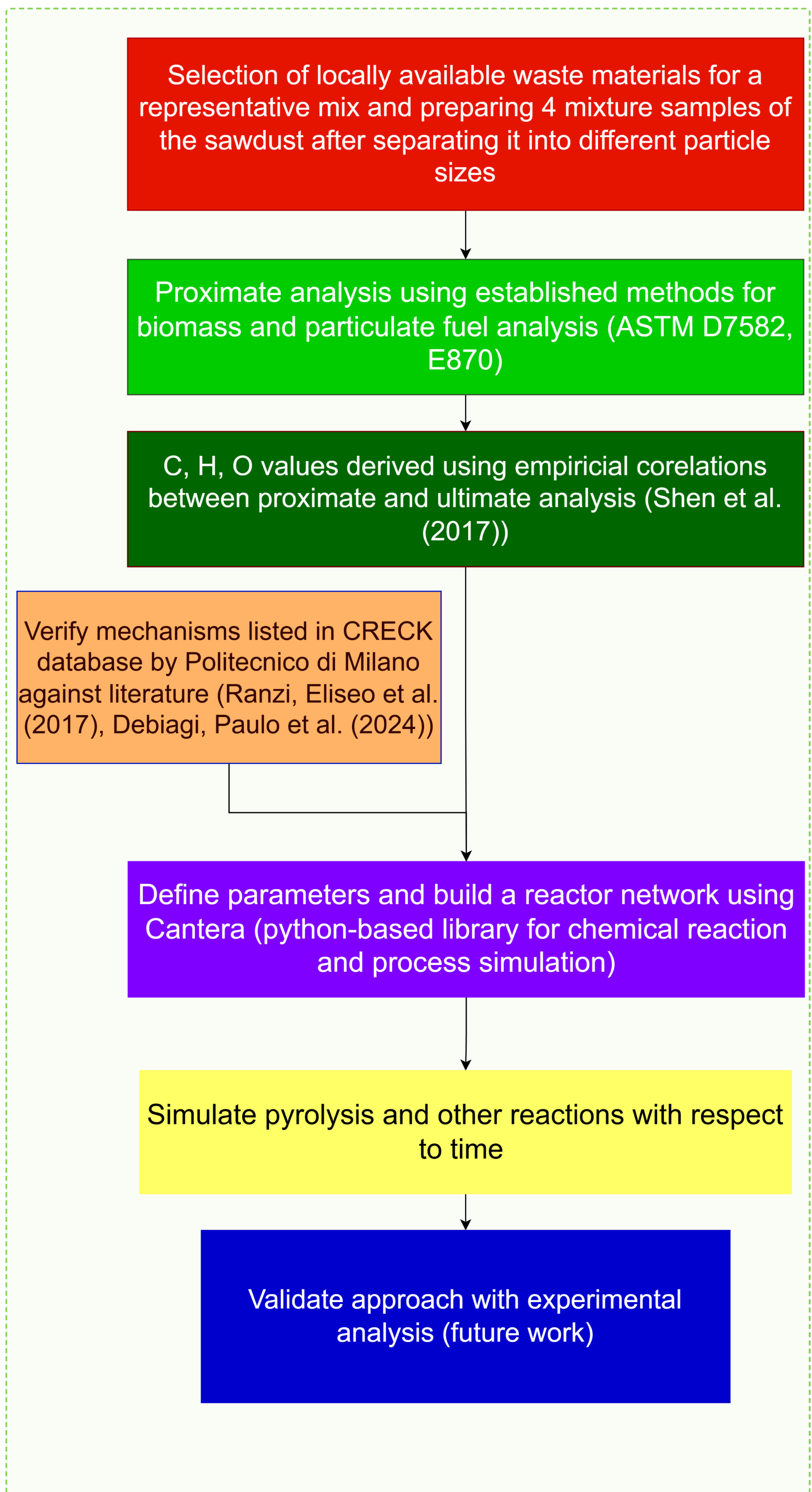

Figure 1. Schematic of the proposed methodology.

04 RESULTS

A model to simulate pyrolysis reactions was built and has been discussed below. The plots show the temperature rise and evolution of volatile content from the feedstock in the coarse-heavy and fine-heavy cases. The fine-heavy sample shows relatively fast devolatilization than the coarser counterpart.

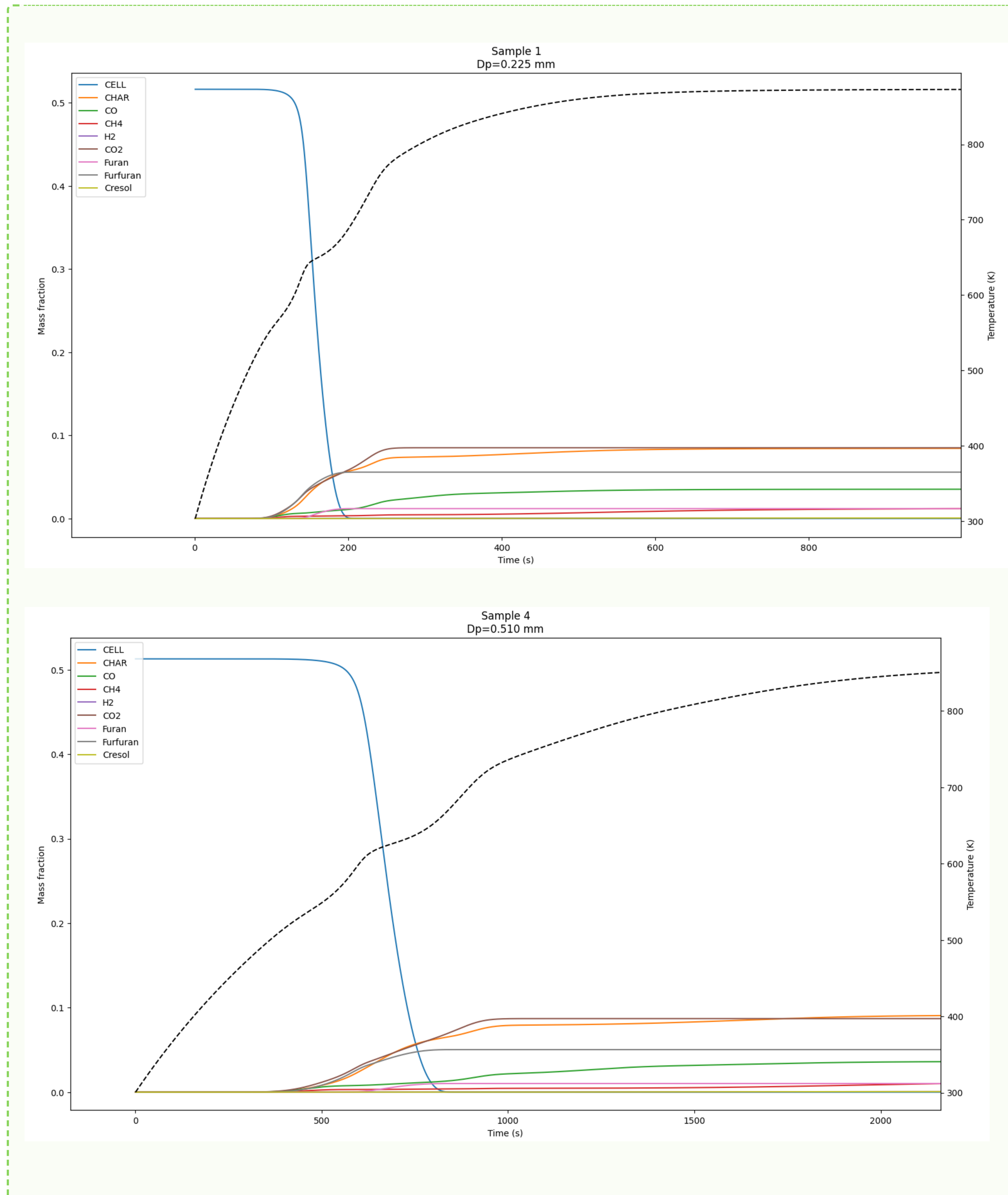

Figure 2. Evolution of species from the pyrolysis of finest versus coarsest sample over t=3600 s

Sample no.	Coarse flakes (sieve size 4.75 mm, values in g)	Semi fine (sieve size 600 micron, values in g)	Fine (sieve size 150 micron, values in g)
1	1	2	2
2	1.6665	1.6665	1.667
3	2.5	1.5	1
4	2.75	1.25	1

Table 1. Composition of sawdust samples used in the study.

Sample	Moisture Content	VM	Ash	FC
1	7.21	85.90	1.48	5.41
2	7.10	86.99	1.62	4.30
3	7.16	88.23	1.76	2.85
4	7.36	84.93	2.19	5.51

Table 2. Proximate analysis of sawdust samples

Image 1. Sawdust samples (coarse and semi-fine) being used for study, mixture samples after Volatile Matter (VM) content test

05 DISCUSSION

Smaller particles exhibited faster heating and accelerated cellulose conversion due to enhanced heat transfer. The predicted trends demonstrate the coupling between particle-scale thermal behavior and temperature-dependent pyrolysis kinetics.

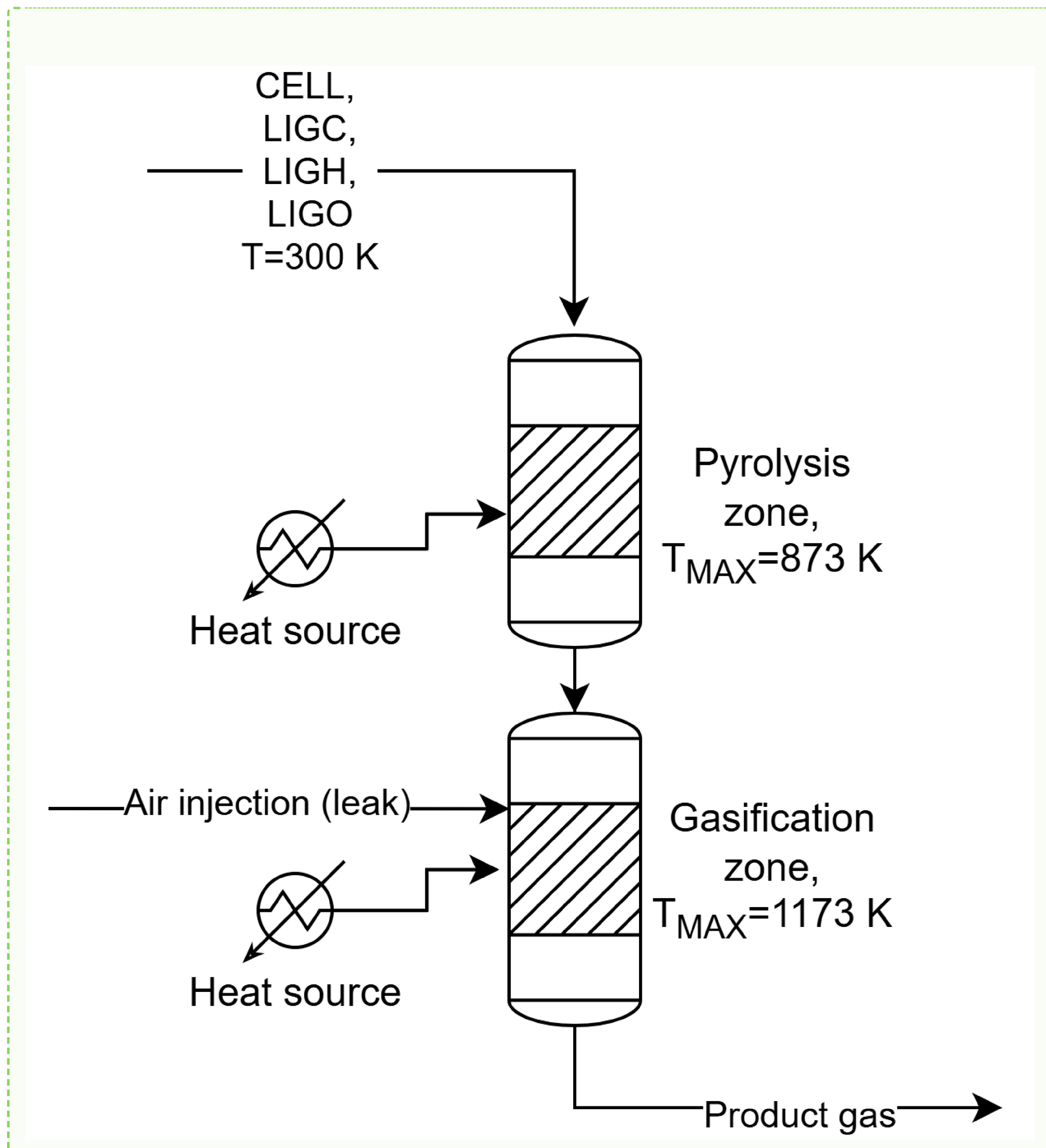

Figure 3. (Above) Schematic for the Reactor model **(Below)** Representative image of the experimental reactor

06 CONCLUSIONS

- A Cantera-based reactor framework was developed to investigate particle-size effects on biomass pyrolysis behaviour. The model successfully captured the evolution of key pyrolysis products, including char, CO, CH_4, and H_2, across different particle sizes, and showed the impact of size on the rate of evolution of different products.

> *Particle sizes affect the rate of reaction, and could also affect the overall efficiency of Pyro gasification systems, making size-reduction important.*
> — KEY FINDING

07 FUTURE WORK

Future work includes testing model with improved gasification mechanisms, incorporating air gasifying conditions, analysing constituent ratio impacts and validating results against experimental results

08 REFERENCES

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